

Datasets included in BEAR

Appendix

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BEAR is available at <https://github.com/wwiecek/BEAR>. For those interested in accessing data for individual datasets (which often include additional variables and structure), BEAR stores each processed dataset as a separate `.rds` file under `data/`; dataset processing is carried out under `process/`. Raw source data are not distributed with the repository, but their source, availability, and relevant processing decisions are described below.

Arel-Bundock et al 2022 (political science)

Reference: Arel-Bundock et al. (2022).

Research question: assess statistical power in political science research

Data availability: replication package (zip file with date 20241010) can be found at <https://osf.io/fgdet>. All data are in public domain.

Data description and source: Authors searched for (and emailed authors of) meta-analysis articles across 141 journals in political science; this led to “a dataset of 16,649 hypothesis tests, grouped in 351 meta-analyses, reported in 46 peer-reviewed meta-analytic articles”

Notes: There is possibly a considerable overlap with Askarov et al dataset, but these two datasets are worth analysing separately given their different focus.

Data processing: using the same code as original authors (replication package that runs on `makefiles`), but skipping some additional processing that they’ve done for their paper. No manipulation of data was done when creating BEAR dataset. We verified that the distributions of z-values matched.

Askarov et al 2023 (economics)

Reference: Askarov et al. (2023).

Research question: Impact of mandatory data sharing on (excessive) statistical significance in economics papers

Data availability: file `Mandatory data-sharing 30 Aug 2022.dta` may be downloaded from <https://github.com/anthonydouc/Datasharing/blob/master/Stata/>; reproducibility package at <https://dataverse.harvard.edu/file.xhtml?fileId=7884702&version=1.0>

Data description and source: The paper includes 64,000 economic estimates surveyed by Ioannidis et al. (2017) plus results of “searching through numerous databases, journal websites, and also an email survey of 109 authors known to have produced meta-analyses of economic topics. We cast the widest net possible to capture empirical economic estimates. Thus, some of the papers included in these meta-analyses were originally published in political science and psychology journals. Nevertheless, these meta-studies deal with economics topics.” Later they note: “we collect 166,924 parameter estimates from 345 distinct research areas, of which 20,121 were published in 24 of the leading general interest and field journals” The paper mentions 12,521 observations before and 2,537 after data sharing policies in 24 journals; reproducibility dataset has 32 journals (authors removed some smaller ones) and 22,172 rows. The rows are meta-analysis estimate rows, not all results from each paper: the meta-analyses collect comparable effect sizes for specific economic phenomena and are not used only to identify papers for full extraction.

Data processing: We use the same source file `Mandatory data-sharing 30 Aug 2022.dta` (11.6 MB) as in the reproducibility package. No additional manipulation of data was done when creating BEAR dataset.

Barnett and Wren

Reference: Barnett and Wren (2019).

Research question: Bias for statistical significance in health and medical journals

Data availability: The dataset `Georgescu.Wren.RData` may be downloaded from <https://github.com/agbarnett/intervals/tree/master/data> and also <https://github.com/jdwren/ASEC>. Data file is `Georgescu.Wren.RData`. Availability: no clear licence given in the repo but the article is CC BY 4.0 in BMJ Open.

Data description and source: The dataset is a collection of confidence intervals for ratio estimates from Medline papers from 1976-2019. Authors scraped (via regular expressions, followed by an independent check using another data mining algorithm, manual checks for 10,000 intervals) 968,000 CIs from abstracts and 350,000 from full texts.

Notes: These are “ratio estimates” like odds ratios, hazard ratios and risk ratios. Note that binary outcomes tend to have (much) lower information than continuous outcomes.

Data processing: We excluded a small fraction of data. First we restricted to 95% confidence intervals only: where `ci.level` was missing we assumed it was 0.95; among intervals with a known `ci.level`, about 0.3% were not 95% and were dropped. We dropped intervals with non-positive width (i.e. `lower >= upper`). We used log scale (we replaced zero or negative lower bounds with a small positive constant; this affected about 0.1% of the sample) and backed out an approximate standard error and point estimate under the usual normal approximation for a 95% CI, setting point estimate to half-point of the interval.

Bartoš et al. 2025 (exercise)

Reference: Bartoš et al. (2025), which builds on another analysis, Singh et al. (2025).

Research question: What is the effect of physical exercise on cognition, memory, and executive function; with extra focus on selective reporting and heterogeneity.

Data availability: Replication data are publicly available via PsyArXiv https://osf.io/preprints/psyarxiv/qv8e2_v1 under Creative Commons By Attribution 4.0 license.

Data description and source: “2,239 effect-size estimates from 215 meta-analyses of randomized controlled trials” was done by Bartoš et al (2025) based on extension of work by Singh et al (2025)

Data processing: z-values were computed as effect size divided by its reported standard error. No filtering or recoding beyond variable renaming was required.

Additional variables used: None.

Brodeur et al. 2024 (economics)

Reference: Brodeur et al. (2024).

Research question: are preregistration and pre-analysis plans associated with reduced p-hacking and publication bias in economics?

Data collection: manually extracted test statistics from articles identified as RCT studies in leading economics journals published 2018-2021. Extraction was performed table by table. They collected coefficients of interest from results tables, excluding constants, balance checks, robustness checks, regression controls, and placebo tests. Each source row corresponds to a single reported test statistic, linked to a paper-level identifier. Unlike most other sources in BEAR, this means the source data have dozens of estimates per paper: 314 articles with 16,390 estimates. After BEAR processing and filtering, we retain 15,917 RCT estimates.

Data availability: replication data file `merged.dta` may be downloaded from <https://dataverse.harvard.edu/file.xhtml?fileId=7884702&version=1.0>

Data processing: the authors' reported z-statistics were used directly and the reported coefficients and standard errors were retained. We retain RCT rows and use the source `zstat` field because the alternative `myz` field is not populated for all publication years in the replication file. We also retain a grouping variable distinguishing not preregistered studies, preregistered studies without a PAP, and preregistered studies with a PAP.

Lang 2025 (economics)

Reference: Lang (2025).

Research question: how frequently are statistically significant rejections of nulls in economics false positives?

Data collection: This paper uses empirical top-five economics papers published in 2021 and Brodeur, Cook, and Heyes dataset of 2015 and 2018 articles in 25 leading economics journals. In each paper, research assistants checked abstracts and results text, kept estimates emphasized in abstracts or as principal results, used the authors' preferred specification when several specifications tested the same hypothesis, and excluded robustness checks or heterogeneity analyses unless heterogeneity was central to the article.

The source data contain 3,885 extracted hypothesis tests from 736 papers. The main sample in the Lang paper is the subset of 2,082 tests with $1.96 \leq |t| < 10$. The paper also reports a one-null-per-article: 736 t-values, but without row or paper identifiers. We do not use the latter.

Data processing: We use the full 3,885-test source table. We use supplied t-statistics (derounded by Lang) as the `z` value and compute two-sided p-values from $|z|$. We add source-article DOIs based on bibliographic metadata matching, and the reported or derounded coefficients and standard errors. The one-null-per-article t-vector is not represented in the row-level BEAR data because its source file has no identifiers that would support matching to the extracted-test table.

Notes: Presumably there is a strong overlap with Brodeur in terms of selected papers, but with differences in extracted values.

Chavalarias et al. 2016 (MEDLINE/PubMed)

Reference: Chavalarias et al. (2016).

Research question: reporting of p-values in the biomedical literature

Data collection: the authors used large-scale text mining of MEDLINE abstracts and PubMed full texts: 4.5mln p-values in 1.6mln MEDLINE abstracts; 3.4mln p-values in 385k PubMed full-text articles.

Data availability: the extracted p-value dataset is publicly available <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/6FMTT3> (about 6GB of data) As per the hosting platform: this dataset is released under the Creative Commons Attribution-NonCommercial 4.0 International License.

Data processing: z-values were derived from p-values assuming two-sided tests. P-value truncation operators were processed by collapsing variants such as <<, <<<, <=, *less than*, and =< into <. We dropped 0.7% of rows where p-values did not have a “plain” format and 0.08% of rows where truncation could not be unambiguously classified.

Additional variables used: source indicator (abstract versus full text).

clinicaltrials.gov snapshot

Trials which reported results at clinicaltrials.gov (National Library of Medicine 2025)

Snapshot date: 12 August 2025

Data source: This is a set of flat file of studies obtained via <https://aact.ctti-clinicaltrials.org/> (over 2GBs of data for relevant tables), with relevant columns extracted and joined by us before BEAR processing.

Data availability: terms and conditions for ClinicalTrials.gov are at <https://clinicaltrials.gov/about-site/terms-conditions>. We make distributions of derived effect sizes available in accordance with ToC’s. We only use publicly available data and do not imply any endorsement of this work by National Library of Medicine, ClinicalTrials.gov or AACT.

Data processing: extensive. We kept only studies where status was “Completed” and only rows of data where outcome type was “Primary”. We set year to the year of `completion_date`.

See *Standard procedure for dealing with p-values and confidence intervals* below for how we created and chose the z-values based on p-values and confidence intervals. Briefly, we calculated z-values using two separate procedures (based on p-values and confidence intervals) and, if both were available, we made a judgement on which one was more appropriate using a pre-defined rule. ClinicalTrials.gov reports the number of sides for confidence intervals but not for p-values, so we treated p-values as two-sided. We dropped rows only when `z` was missing. Notably, infinite z-values can arise when `p_value` is recorded as exactly 0, and these are retained because they are not missing.

Cochrane Database of Systematic Reviews (CDSR)

Trials with data available at CDSR, <https://www.cochranelibrary.com/> (Cochrane Collaboration 2025)

Snapshot date: 20 November 2025

Data collection: using R package `cochrane` (Schwab 2024) we merged data from 8,726 records of reviews in CDSR and merged them into a single file `cdsr_interventions_19nov2025.csv` (63 MB).

Data processing: minimal processing with extensive filtering. Final BEAR effect sizes are recalculated for continuous outcomes as standardised mean differences and for dichotomous outcomes as probit-transformed effects. Only efficacy outcomes are retained. The dataset currently has about 38,000 rows, compared to over 800,000 rows in the original extract.

On processing side, we cleaned up study years, categorised measures (risk ratio, odds ratio, mean difference etc.) and experimental designs, especially an “RCT” flag (based on scanning of abstracts of each review for inclusion criteria). Most studies in CDSR are RCTs but some reviews also allowed quasi-experimental studies, which prevented us from categorising many studies. We also classify outcome rows into **efficacy**, **safety**, **dropouts**, and **bias** based on the comparison, outcome, and subgroup labels.

Unlike in most of the other datasets in BEAR, we filter CDSR data heavily. First, we only use the outcome and comparison of each review that are coded as “1”, as that is most likely the primary outcome and most relevant comparison. Secondly, we keep only rows classified as **efficacy**, excluding likely safety, dropout, and bias-related outcomes based on comparison, outcome, and subgroup labels. Thirdly, we only use studies with continuous and dichotomous outcomes, removing rows where effect estimates are based on instrumental variables or based on individual patient data. Lastly, we remove rows where measure of effect is unknown (retaining: OR, RR, Peto OR, mean difference, standardised mean difference, and risk difference) and rows where there are zero subjects.

Additional grouping variable: we retained a “source data type” variable to distinguish estimates from published, unpublished, “sought”, and mixed data sources. We also retain an **outcome_group** variable in the processed Cochrane data.

Costello and Fox 2022; Yang et al 2023, 2024 (ecology and evolution)

Reference: these two datasets come from the same replication package and shared data processing pipeline in Yang et al. 2024 (Yang et al. 2024). Original references for the datasets are Yang et al. 2023 and Costello and Fox 2022 (Yang et al. 2023; Costello and Fox 2022).

Research question: decline effects in ecology (Costello and Fox) and publication bias and performance of empirical research (Yang et al)

Data collection: Costello and Fox look at 466 meta-analyses obtained through searching Web of Science. Reproducibility dataset includes 88,000 rows among 466 analyses in 232 papers. Subsequently, Yang et al removed duplicated studies and zeroes, which meant discarding about 20% of rows. Yang et al collect 102 meta-analyses published 2010-2019 in relevant journals, de-duplicated against Costello and Fox dataset.

Data availability: two datasets may be downloaded from https://github.com/Yefeng0920/replication_EcoEvo_git/. Available under CC BY 4.0 licence.

Data processing: no additional data processing other than cleaning up study years. Note that the replication package for Yang et al does not include meta-analysis indicators.

Additional variables used: we retained a measure variable (“lnRR”, “SMD”, “Zr” or “uncommon”)

EU Clinical Trials Register (EU CTR)

Trials which reported results at the EU Clinical Trials Register, <https://www.clinicaltrialsregister.eu/> (European Medicines Agency 2025).

Snapshot date: 18 January 2026

Data collection: we used R package `ctrdata` (Herold 2025) to create a snapshot of EU CTR (`collection = "euctr"`); we downloaded trial records with results only. We stored the downloaded records in a local SQLite database and extracted a fixed set of protocol and results fields plus several fields computed by `ctrdata` (trial phase, sample size, and a “first primary endpoint” p-value/size derived by the package). We only use the EUCTR portion of this extraction in BEAR.

Data availability: EUCTR is publicly accessible. The EUCTR legal notice notes that publication on EUCTR does not constitute an endorsement of the information reported.

Data processing: extensive. We use only primary endpoints. We defaulted missing CI levels to 95% and where truncation of p-values was not stated we assumed equality (the latter was the case for about 15% of data). Where EU CTR reported whether a result was one- or two-sided, we used this information for both confidence intervals and p-values; when it was missing, we assumed two-sided. Rest of the derivation of z-values followed the *Standard procedure for dealing with p-values and confidence intervals* described below. We set `year` to record *entry* date (since “completion date” field was not accessible), dropped rows only when z was not finite, and harmonised phase labels to follow the same naming convention as ClinicalTrials.gov.

Additional variables used: we retained phase labels, a measure class (derived from the estimand label), primary endpoint sample size

Head et al 2015 (PubMed papers)

Reference: Head et al. (2015).

Research question: extent of p-hacking and its impact on meta-analyses.

Data collection: PubMed papers that are open access, up to 2014. Authors used regular expressions to extract p-values from text of Abstract and Results, but not tables. Downloaded data has about 220,000 studies and about 2mln rows.

Data availability: dataset available at <https://datadryad.org/dataset/doi:10.5061/dryad.79d43> under CC0 1.0 Universal licence.

Data processing: we followed the same minimal clean-up steps (e.g. removing values found in large supplementary sections) as the original paper and also attached PubMed IDs to the dataset using DOI look-up.

When constructing the BEAR dataset, we treated p-values recorded as exactly 0 as truncated and set the z-operator to $>$. We recoded all inequalities to sharp inequalities.

Additional variables used: we use the “Abstract vs Results” variable for grouping

Jager and Leek

Reference: Jager and Leek (2014).

Research question: estimating science-wise false discovery rate.

Data collection: authors used a custom program to extract p-values from scraped PubMed abstracts for papers published in 5 main medical journals 2000-2010. 15,653 p-values are available in 5,322 articles.

Data availability: the data file `pvalueData.rda` is available <https://github.com/jtleek/swfdr>. License for the programs in that repository is GNU GPL, although a license for the dataset is not stated, as far as we are aware.

Data processing: there was only minimal processing. We derived z values from p-values assuming they were two-sided.

Notably, large proportion of p-values is truncated, almost always at 0.0001, 0.001, 0.01, or 0.05. As in all datasets, we retained information on truncation. We also treated p-values recorded as exactly 0 as truncated (z-operator $>$). We created a crude flag for RCTs by searching the paper titles for “randomized”, “randomised” and “controlled”.

Metapsy (psychotherapy)

Studies included in *Metapsy* (<https://www.metapsy.org/>), a set of living meta-analytic databases of randomised trials of psychotherapeutic interventions.

Snapshot date: Jan 2026

Data availability: Metapsy is maintained by an international collaboration led by Vrije Universiteit Amsterdam. Databases are available at <https://www.metapsy.org/> and via R package `metapsyData` (Harrer, Karyotaki, and Cuijpers 2022). Metapsy FAQ states that all data provided as part of the project are open-access and requires attribution.

Data processing: we merged 20 individual datasets in Metapsy with some cleaning (e.g. to extract study dates from titles). Almost all estimates are Hedges’ g , but for one of the databases (`total-response`) we converted log-odds ratios to SMD (dividing by $\Pi/\sqrt{3}$ to scale SD of logistic regression).

Open Science Collaboration 2015

Reference: Open Science Collaboration (2015).

Research question: replications of a quasi-random sample of 100 experiments in psychology.

Data collection: authors of the paper make a comprehensive dataset of results available with their paper. In BEAR we did not use the original studies (since all studies in that set have $p < 0.05$), only replications.

Data availability: the data file `rpp_data.csv` is available as part of repository for the paper <https://osf.io/ytpuq/overview> under CC0 1.0 Universal license.

Data processing: we derived z values from the replication p-values assuming two-sided p-values (we use the replication p-value column, not the original-study p-value). Because the replication dataset does not carry direction in a way we use here, z values are unsigned. We treated p-values recorded as exactly 0 as truncated and recorded this as a lower bound on z (z-operator $>$); otherwise we treated p-values as exact (z-operator $=$). We retained the replication sample size as `ss`.

psymetadata and Nuijten et al (psychology datasets)

Reference: Set of psychology datasets compiled by Rodriguez and Williams (2022). We also disaggregate a large meta-meta-analysis of intelligence research (Nuijten et al. 2020).

Snapshot date: Jan 2026

Research question: *psymetadata* is an R package that curates multiple open datasets from published meta-analyses in psychology for teaching and demonstrations. Nuijten et al. (2020) is a large meta-meta-analysis of intelligence research, which focused on power and small study effects.

Data availability: 22 datasets are distributed with the *psymetadata* R package (CRAN). We do not use its Many Labs 2 dataset, which we obtain from the original source and include separately. After excluding Many Labs 2 and separately representing the intelligence meta-meta-analysis, the remaining psychology datasets vary from 65 rows of data to over 1,000. The package is GPL-3; individual datasets originate from the cited source papers.

Data processing: minimal. We used regular expressions to extract year labels from titles. We dropped some invalid rows (missing, non-finite, or non-positive standard errors).

Additional grouping variable: name of the original *psymetadata* dataset.

Many Labs 2

Reference: Klein et al. (2018).

Snapshot date: Jan 2026

Research question: replications of 28 classic and contemporary effects in psychology, with the same effects tested across many samples and research sites.

Data availability: the authors make their replication package available through OSF. We use the paper-included Many Labs 2 analyses from the key table and the site-level aggregated replication results from the public data package.

Data processing: we retained the 28 preregistered analyses included in the paper. Each row is a site-level replication estimate for one analysis, so the dataset has many more rows than analyses. We use the Many Labs 2 analysis identifier as *metaid*. Thus, a single finding such as *Kay.1* contributes one BEAR row for each sample/site estimate, rather than one row for each alternative statistical scale reported for the same sample. The additional grouping variable distinguishes analyses designated as primary or secondary by the original authors.

The retained effect-size measure is the correlation scale. We used the reported *r* estimate as *b*, its reported variance to compute *se* where available, and then computed $z = b / se$. For two analyses where the variance of *r* was not available but a signed equivalent test statistic was available, we used that statistic as *z* and derived the implied standard error. Original-study effects were matched by analysis where available and stored separately from the replication estimates.

Additional grouping variable: primary or secondary analysis designation.

SCORE

Reference: Tyner et al. (2026).

Research question: investigating the replicability of published claims in the social and behavioural sciences.

Data availability: the SCORE replication package is available at <https://osf.io/g5sny/>. The package includes replication data as well as reviews of claims made in included papers.

Data description and source: BEAR includes two SCORE outputs. The replications source dataset has 548 rows: 274 original claims and 274 matched replications. After BEAR processing and filtering, 267 matched replication rows are retained. The “all claims” source dataset has 3,066 claim-level rows from the set of replicated papers (but no replications of these claims), where we select one statistic per claim; after BEAR processing and filtering, 1,946 claim rows are retained.

Data processing: For the matched replication output, original and replication statistics we used the package’s converted correlation scale where available. For the all-claims set, effects are heterogeneous across the source papers.

When several inputs are available for z-value construction in the all-claims dataset, we prefer reported z, then reported t, then coefficient divided by standard error, then signed square-root F for numerator df 1, then a 95% confidence interval with a point estimate, then a two-sided p-value conversion. For the all-claims text, it is typical to have multiple statistics backing up a single claim, e.g. “ $F(1,38) = 3.73$, $p = .033$, partial eta-squared = .16; $F(1,39) = 7.28$, $p = .010$, partial eta-squared = .16; $F < 1$ ”. We created a rule to pick one statistic per claim using agreement with the SCORE significant/nonsignificant coding, statistic provenance, exactness, available effect and standard error, and text order.

Additional grouping variables: discipline and SCORE source indicator.

What Works Clearinghouse (education)

Large database of education effect sizes created by Institute of Education Sciences (Institute of Education Sciences 2025).

Snapshot date: 17 May 2025

Data availability: flat files are available at <https://ies.ed.gov/ncee/wwc/studyfindings>. Per permissions and disclaimers section on that website: “Unless stated otherwise, all information on the U.S. Department of Education’s IES website ... is in the public domain and may be reproduced, published, linked to, or otherwise used without permission from IES.”

Data collection: The WWC source data have 13,054 findings (data rows) across 1,908 reviews. The dataset includes effect sizes, but not standard errors, as well as p-values reported by the studies and p-values re-calculated by the WWC reviewers. After BEAR processing and filtering, 12,045 findings are retained.

Data processing: We retained only RCTs and quasi-experimental studies (98% of data altogether). Where available, we preferred effect sizes and p-values calculated by WWC to the ones reported by the studies. We replaced p-values that were equal to zero (this occurred in 3% of the dataset) with truncated p-value (i.e. assumed $p < 10^{-16}$). We calculated z-values from these p-values under assumption of two-sided tests.

Additional grouping variable: “outcome domain” variable (e.g. “academic achievement”, “alphabets”)

Standard procedure for dealing with p-values and confidence intervals

Most datasets come with pre-computed effect sizes and standard errors. Some (especially databases of clinical trials and large sets of scraped studies) report p-values. Reporting in clinical trial database is done by investigators and data entry fields are not standardised, requiring more extensive clean-up and a procedure for deriving z-values.

Cleaning up data

For confidence interval levels, we clean them up by setting values above 100 or below 0 to missing, and treating values below 1 as proportions rather than percentages by multiplying by 100. This means that, for example, 0.95 is treated as a 95% confidence interval. For missing values in EU CTR database, we assume 95%.

In clinical trial datasets we classify the effect measure using text matching to a small set of classes (Mean Difference, Odds Ratio, Risk Ratio, Hazard Ratio, Geometric Ratio, Risk Difference, Difference in Percentages, Ratio/Other Ratio, and Other). This string-based mapping is an arbitrary choice. Unstandardised data in these datasets have many hundreds of values.

Confidence intervals

When working with CIs, ratio measures were analysed on the log scale (if and only if lower bound, upper bound, and point estimate were all strictly positive) and non-ratio measures were always kept on the raw scale.

For z-statistics derived from confidence intervals, we computed the critical value using stated CI percent and the reported number of sides (if missing, we defaulted to a two-sided interval). To avoid undefined critical values (e.g. when at 100% or extremely close to it), we bounded the implied α at a small value. We then computed two one-sided standard error estimates from the upper and lower bounds and used their average when both were finite.

We computed a simple symmetry diagnostic for the confidence interval on the chosen scale, defined as the smaller of the two half-widths divided by the larger. We treated intervals as “symmetric enough” only if this ratio exceeded 0.8. This threshold is arbitrary and affects when we trust the CI-derived z-statistic.

We also flagged potentially non-Wald intervals using text matching on type of parameter column, if available, searching for keywords such as median, Hodges, posterior/Bayes, exact, Fieller, bootstrap, and permutation. Any row flagged this way was treated as unreliable for CI-based inference, regardless of its numeric properties.

p-values

The default approach, used for several datasets which report p-values, especially those scraping data from PubMed/Medline, is to treat p-values as two-sided ($z \leftarrow \text{qnorm}(1 - p/2)$). In EU CTR, when the result is marked as one-sided, we use $z \leftarrow \text{qnorm}(1 - p)$ instead; missing sidedness is

treated as two-sided. ClinicalTrials.gov does not provide a separate p-value sidedness field, so its p-values are treated as two-sided.

For clinical trials databases (clinicaltrials.gov and EU CTR) we sometimes found ambiguity in how p-values were recorded. Sometimes significant results report p-values close to 1, suggesting that authors reported cumulative distribution, $\Phi(z)$. Therefore for each row we also calculated $2 \cdot \min\{p, 1 - p\}$. When a CI-derived z-statistic was available, we chose whichever p-value led to a z-value that was closest to the CI-derived z.

To avoid dropping rows, we retained unsigned z-statistics when the sign could not be determined. If the sign of effect was known, we applied it to the “p-derived” z; otherwise we used absolute value.

Choice of a single z-value when multiple candidates available

In situations where z can be derived from both CIs and p-values. We choose a single “best” z-statistic per row as follows. We prefer the CI-derived z when it looked like a plausible Wald interval on the chosen scale, meaning: (1) finite z and standard error, (2) positive standard error, (3) not flagged as non-Wald by keyword search in measure label, and (4) either missing symmetry information or symmetry above 0.8. Otherwise we used the p-derived z .

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